

Comparative Analysis of Convolutional Neural Network Depth in Multi-Class Recyclable Waste Classification

Roger Arias¹, Irene Ramirez¹, Steven Weber¹

¹University of North Florida

n01500586@unf.edu, n01512644@unf.edu, n01396532@unf.edu

Abstract

This paper investigates the impact of architectural depth on the classification of recyclable materials—specifically glass, metal, paper, and plastic—using Convolutional Neural Networks (CNNs). We designed and evaluated two models: Model A, a baseline 2-layer CNN, and Model B, a 3-layer CNN with increased filter depth. Using a dataset of 12,875 images, we found that increasing architectural depth and filter density improved overall test accuracy from 49.20% to 51.22%. Notably, the deeper architecture showed a specialized improvement in the classification of "Paper" (+4.14%) and "Plastic" (+4.10%), suggesting that additional layers are critical for capturing the complex, non-uniform textures inherent in deformable recyclable materials.

Code - https://github.com/weberstevnn/AI_Project4

Datasets - <http://ailab.cceec.unf.edu/recycledata.html>

1. Introduction

The global escalation of waste production has made efficient recycling a cornerstone of environmental sustainability. However, the efficacy of recycling programs is often hindered by the logistical challenge of waste sorting. Traditional manual sorting is labor-intensive, hazardous, and economically inefficient. While automated systems exist, they often rely on expensive sensors or weight-based mechanics that struggle with the increasing variety of modern consumer packaging.

1.1 The Computer Vision Challenge

The primary motivation of this project is to explore the viability of a low-cost, image-based classification system using Convolutional Neural Networks (CNNs). In the context of computer vision, the "Waste Classification Problem" is uniquely difficult for several reasons:

- **Intra-class Variation:** A "plastic" item could be a clear water bottle, a red detergent jug, or a crumpled white film. These objects share a class label but possess almost no shared color or shape features.

- **Inter-class Similarity:** At lower resolutions, the visual signature of clear glass and clear plastic is nearly identical, leading to high rates of model confusion.
- **Deformability:** Unlike rigid objects (like cars or furniture), materials like paper and plastic are "deformable." Their geometric features change entirely when they are crushed, folded, or torn.

1.2 Project Objective

The goal of this research is to investigate how architectural depth affects a model's ability to extract these complex features. By comparing a baseline 2-layer CNN (Model A) with an expanded 3-layer CNN (Model B), we aim to determine if additional convolutional layers provide the hierarchical feature extraction necessary to distinguish between deformable materials. This study serves as a proof-of-concept for lightweight, deployable AI models that can be integrated into "smart" recycling bins or local waste-processing facilities.

2. Related Work

The automation of waste sorting has been a subject of significant research in the field of computer vision, evolving from traditional feature-engineering techniques to modern deep learning architectures.

2.1 Traditional vs. Deep Learning Approaches

Prior to the widespread adoption of Convolutional Neural Networks (CNNs), waste classification relied heavily on manual feature extraction methods, such as color histograms and Scale-Invariant Feature Transform (SIFT). While these methods were computationally lightweight, they lacked the robustness to handle the high intra-class variation found in recyclables—where a single category like "plastic" can encompass a vast array of shapes, colors, and opacities. Recent literature confirms that CNNs have superseded these methods by allowing for hierarchical feature extraction directly from raw pixel data, which is essential for identifying materials in unconstrained environments (LeCun, Bengio, and Hinton 2015).

2.2 Architectural Depth in Waste Classification

A central theme in contemporary AI research is the trade-off between architectural depth and classification accuracy. Established models such as ResNet and VGG have demonstrated that increasing the number of convolutional layers allows a network to learn increasingly abstract features. However, these models often require massive datasets and significant computational power. In the domain of waste classification, research using the "TrashNet" benchmark (Yang and Thung 2016) has shown that while deeper networks generally improve accuracy, they also face diminishing returns and potential overfitting when applied to specific, localized datasets.

2.3 The CCEC AI Lab Benchmark

This study builds upon these established principles by utilizing the specialized dataset provided via the UNF College of Computing, Engineering, and Construction (CCEC) Artificial Intelligence Laboratory (<http://ailab.ccec.unf.edu/recycledata.html>). This dataset provides 150x150 colored images across four material classes, offering a standardized environment to test architectural efficiency. Our work specifically isolates the impact of a single-layer depth increase (moving from a 2-layer to a 3-layer CNN) to determine the minimum architectural complexity required to improve the classification of "deformable" materials like paper and plastic, which remain a primary challenge in automated sorting.

3. Methodologies

The research objective was to isolate the effect of architectural depth on classification performance. This section details the data pipeline, the specific configurations of the two architectures, and the training protocol.

3.1 Data Pipeline and Preprocessing

The dataset was retrieved from the UNF CCEC AI Lab repository. To ensure computational efficiency, all input images were standardized to a 150x150 pixel resolution while maintaining the full 3-channel RGB color space. This preservation of color data is vital for distinguishing between reflective surfaces (glass and metal) and matte textures (paper). The total dataset of 12,875 images was partitioned into 8,641 training samples, 2,104 validation samples, and 2,130 test samples, providing a robust foundation for evaluating the models' generalization capabilities on unseen data.

3.2 Architecture Design

We designed two distinct Convolutional Neural Network (CNN) architectures to isolate the impact of convolutional depth. Model A serves as the baseline, utilizing a shallow 2-layer configuration, while Model B represents the experimental deeper architecture. Their specific configurations are detailed in Table 1.

Table 1: Comparative Architecture Specifications

Layer Type	Model A	Model B
Conv_1	16 Fil.	32 Fil.
Conv_2	32 Fil.	64 Fil.
Conv_3	-	128 Fil.
Pooling	Max	Max
Activ.	ReLU	ReLU
Optim.	Adam	Adam

Note: "Fil." Refers to the number of convolutional filters.

3.3 Training and Optimization

Training was conducted over a 10-epoch cycle utilizing the Adam optimizer with a learning rate of 0.001 and Cross-Entropy Loss. A significant technical challenge involved the transition between development on Apple Silicon (MPS) and deployment on Windows-based CPU or CUDA environments.

This challenge was mitigated by implementing a dynamic hardware-agnostic detection strategy. Rather than hard-coding device parameters, the system executes a runtime check to prioritize available hardware acceleration (MPS or CUDA) before falling back to the CPU backend. This ensures consistent tensor operations and weight initialization across disparate research environments, allowing the model to be trained on high-performance laptops and evaluated on standard laboratory workstations without code modification.

3.4 Automated Experimental Framework

To ensure reproducibility and rigorous comparison, the research was executed through a centralized automation script, `master_run.py`. This framework manages the entire lifecycle of the experiment:

- **Dynamic Backend Initialization:** The system utilizes a runtime check to prioritize Apple Silicon (MPS) for hardware acceleration, falling back to CPU when necessary to ensure cross-platform compatibility.
- **Integrated Evaluation Pipeline:** Beyond standard accuracy, the script generates a comprehensive Classification Report (Precision, Recall, F1-Score) and a Confusion Matrix for each model.
- **Weight Serialization:** Upon completion of the 10-epoch training cycle, the script serializes the model's state dictionary into .pth files, allowing for the "Brain" of the model to be redeployed without retraining.

4. Experiments and Results

4.1 Performance Metrics

The models were evaluated using Test Accuracy, defined as the ratio of correct predictions to the total 2,130 test samples. Both models reached a training plateau at 10 epochs, providing a stable point for a comparative performance audit.

4.2 Comparative Analysis

Model B, utilizing the deeper 3-layer architecture, demonstrated a measurable performance gain over the

Model A baseline. This confirms that additional convolutional depth allows for the extraction of more complex feature hierarchies.

Table 2: Material Class Accuracy (%)

Class	Model A	Model B	Δ
Glass	54.7	55.0	+0.3
Metal	47.5	45.2	-2.3
Paper	64.3	68.4	+4.1
Plastic	28.8	32.9	+4.1
Total	49.20	51.22	+2.02

Note: Δ indicates the percentage point difference

4.3 Material-Specific Findings

The most significant outcome of the research was the performance spike in deformable categories. Model B achieved a +4.1% improvement in both the Paper and Plastic classes. This suggests that the third convolutional layer is better suited for identifying the irregular geometric patterns (folds, wrinkles, and non-linear edges) typical of these materials. Conversely, the slight regression in Metal (-2.3%) suggests that reflective surfaces might suffer from over-parameterization in deeper models or may require specialized data augmentation to improve invariance.

4.4 Error Analysis

A review of the confusion matrix reveals a persistent bottleneck: high inter-class similarity between Plastic and Paper. At a 150×150 resolution, these materials often exhibit nearly identical visual signatures. Specifically, 258 plastic samples were misclassified as paper in Model B. This indicates that while increasing architectural depth improves feature recognition, visual ambiguity at low resolutions remains a primary constraint.

4.5 Qualitative Discussion

The persistent bottleneck in classifying Plastic (32.9% accuracy) relative to Paper (68.4% accuracy) highlights a fundamental challenge in low-resolution waste sorting. While the 3-layer architecture improved feature recognition for deformable items by over 4%, the model frequently struggled with "Clear Plastic". Under the 150×150 resolution constraint, the lack of distinct edge features in transparent plastics makes them visually indistinguishable from background noise or white paper surfaces. This suggests that while increasing depth to three layers is a successful step for texture recognition, further gains may require Data Augmentation techniques like color jittering to artificially enhance contrast for transparent materials.

5. Conclusion

This study successfully evaluated the impact of convolutional depth on the classification of recyclable materials using the UNF CCEC AI Lab dataset. The experimental results confirm that a 3-layer architecture (Model B) provides a measurable accuracy improvement of 2.02% over a 2-layer baseline, specifically excelling in the identification of deformable materials

like paper and plastic. While visual similarities between classes at 150×150 resolution remain a challenge, this research provides a clear foundation for future work in automated waste sorting, suggesting that architectural depth is a key factor in handling high intra-class variation.

6. References

- AILab UNF. 2026. Recyclable Material Classification Dataset. College of Computing, Engineering, and Construction. <http://ailab.ccec.unf.edu/recycledata.html>.
- LeCun, Y.; Bengio, Y.; and Hinton, G. 2015. Deep Learning. *Nature* 521(7553): 436–444.
- Kingma, D. P., and Ba, J. 2014. Adam: A Method for Stochastic Optimization. arXiv preprint arXiv:1412.6980.